

Barra China A Total Market Equity Model for Long-Term Investors and Barra China Equity Model (CNE5) Comparison Datasheet

August 2018

Features in Barra China A Total Market Equity Model for Long-Term Investors (CNLT)

Robust specific risk model that combines structural and time series components, based on daily asset-level returns.

Broad coverage of the China domestic market including Stock Connect listings and full coverage of major indexes.

Enhanced style factors based on Systematic Equity Strategies.

Separation of industry and market effects through a Market factor improves forecasting accuracy.

Point-in-time fundamental data to reduce forward bias in backtesting.

Enhanced beta estimation with Bayesian shrinkage towards industry betas.

Volatility Regime Adjustment for more timely and accurate response to sudden changes in market volatility.

Optimization Bias Adjustment to reduce biases in risk forecasts for optimized portfolios.

Alignment of the factor structure and responsiveness with an investment horizon of over six months. Available in Stable and Responsive variants.

Full daily history and updates of the model.

CNE5 Model Summary	China A Long-Term Equity Model Summary
Asset Coverage: 2,200+	Asset Coverage: 5489 ¹
Style Factors: 10	Style Factors: 16
Industry Factors: 32	Industry Factors: 32
Covariance Matrix built on daily data	Covariance Matrix built on daily data

Model Feature	CNE5 Model	China A Long-Term Equity Model
Model Start Dates	Barra Aegis: Jan 29, 1999 Barra PortfolioManager: Dec 31, 2002 Barra Models Direct: Jan 29, 1999	Barra Aegis: Dec 31, 1998 Barra PortfolioManager: Dec 31, 1998 Barra Models Direct: Dec 31 1998
Estimation Universe	All China A-Shares (Domestic Equity Assets)	China component of MSCI All Country World Investable Market Index (ACWI IMI) - before June 2008 MSCI China A Onshore IMI - after June 2008
Regression Weighting Scheme	Square root of market capitalization	Square root of market capitalization
Factor Covariance Matrix	Long Half-life for Volatility: 252 days Half-life for Correlations: 504 days Half-life for Volatility Regime Adjustment: 168 days Short Half-life for Volatility: 84 days Half-life for Correlations: 504 days Half-life for Volatility Regime Adjustment: 42 days	Stable Variant Half-life for Volatility: 252 days Half-life for Correlations: 756 days Volatility Regime Adjustment: 84 days Responsive Variant Half-life for Volatility: 84 days Half-life for Correlations: 504 days Volatility Regime Adjustment: 21 days
Specific Risk	Half-life Long Variant: 252 days Half-life Short Variant: 84 days Bayesian Shrinkage is applied	Half-life Stable Variant: 252 days Half-life Responsive Variant: 84 days Bayesian Shrinkage is applied

¹ As of May 2, 2018

Model Feature	CNE5 Model	China A Long-Term Equity Model
	Volatility Regime Adjustment is applied Linked Specific Risk available	Volatility Regime Adjustment is applied Linked Specific Risk available
Industry Allocation Scheme	Single-industry allocation	Single-industry allocation
Source of Industry Scheme	Based on Global Industry Classification Standard (GICS®)	Based on Global Industry Classification Standard (GICS®)

Style Factors	CNE5 Model	China A Long-Term Equity Model
Beta	Captures the market risk that cannot be explained by the Country factor. Historical Beta to market return	Explains common variations in stock returns due to different stock sensitivities to market or systematic risk that cannot be explained by the Country factor Historical Beta
Book-to-Price	Explains the return component attributable to a stock's book-to-price ratio and can be an indicator of value. Book-to-Price	Calculated as the last reported book value of common equity divided by current market capitalization. Book-to-Price
Dividend Yield	N/A	Captures differences in stock returns attributable to stock's trailing 12-month and predicted dividend-to-price ratios - Analyst Predicted Dividend-to-Price - Dividend-to-Price
Earnings Quality	N/A	Explains stock return differences due to uncertainty around company operating fundamentals (sales, earnings, cash flows) and the accrual components of their earnings - Accruals - Balance Sheet Version - Accruals - Cash flow Statement Version
Earnings Variability	N/A	Explains stock return differences due to variability in earnings and cash flows using both historical measures and analysts' forecasts - Standard Deviation of Analyst Forecast Earnings-to-Price - Variability in Cash flows

Style Factors	CNE5 Model	China A Long-Term Equity Model
		• Variability in Earnings • Variability in Sales
Earnings Yield	Describes return differences based on a company's earnings relative to its price. • Analyst Predicted Earnings-to-Price • Trailing Annual Earnings-to-Price • Cash Earnings-to-Price	Describes stock return differences due to various ratios of the company's earnings relative to its price • Analyst Predicted Earnings-to-Price • Cash Earnings-to-Price • Earnings-to-Price • Enterprise Multiple (EBIT to EV)
Growth	Describes return differences of stocks based on their prospects for sales or earnings growth. • Sales Growth • Analyst-predicted Earnings Growth • Earnings Growth Rate Long Term • Earnings Growth Rate Short Term	Measures company growth prospects using sales growth and earnings growth over trailing five years and predicted earnings growth • Analyst Predicted Earnings Long-term Growth • Earnings per Share Growth Rate • Sales per Share Growth Rate
Investment Quality	N/A	Combination of asset, investment, net issuance growth measures that captures common variation in stock returns of companies experiencing rapid growth or contraction of assets • Capital Expenditure Growth • Issuance Growth • Total Assets Growth Rate
Leverage	Captures the return differences between high-leverage and low-leverage stocks. • Market Leverage • Debt-to-Assets Ratio • Book Leverage	Captures common variation in stock returns due to differences in the level of company leverage • Book Leverage • Debt-to-Assets • Market Leverage
Liquidity	Describes the return differences of stocks based on their relative trading activity. • Monthly Share Turnover • Quarterly Share Turnover • Annual Share Turnover	Captures common variations in stock returns due to the amount of relative trading and differences in the impact of trading on stock returns • Annual Share Turnover • Annual Traded Value Ratio • Monthly Share Turnover

Style Factors	CNE5 Model	China A Long-Term Equity Model
		Quarterly Share Turnover
Long-Term Reversal	N/A	Explains common variation in returns related to a long-term (five years ex. recent thirteen months) stock price behavior. It is orthogonalized with respect to the Momentum factor. - Long-term Historical Alpha - Long-term Relative Strength
Mid Capitalization	N/A	Captures non-linearities in the payoff to the Size factor across the market-cap spectrum. This factor is based on a single raw descriptor: the cube of the Size exposure. However, because this raw descriptor is highly collinear with the Size factor, it is orthogonalized with respect to the Size factor. Cube of Size Exposure
Momentum	Explains the return differences of stocks based on their relative performance over the trailing 6-12 months. Relative Strength	Explains common variation in stock returns based on their performance over the trailing 12 months. - Historical Alpha - Relative Strength 12-month
Non-linear Size	Describes size-related returns to stocks that are not captured by the Size factor. Non-linear Size	N/A
Profitability	N/A	Combination of profitability measures that characterizes efficiency of a firm's operations and total activities - Asset Turnover - Gross Profit Margin - Gross Profitability - Return on Assets
Residual Volatility	Explains returns associated with high-volatility stocks that are not captured by the Beta factor. - Daily Standard Deviation Calculation - Cumulative Range - Historical Sigma	Captures relative volatility in stock returns. Consists of three descriptors: (a) the volatility of daily excess returns, (b) the volatility of daily residual returns, and (c) the cumulative range of the stock over the last 12 months. Since these descriptors tend to be highly collinear with the Beta factor, the Residual Volatility factor is orthogonalized with respect to the Beta, Size,

Style Factors	CNE5 Model	China A Long-Term Equity Model
		and Liquidity factors. - Cumulative Range - Daily Standard Deviation - Historical Sigma
Size	Captures the return differences between large-cap stocks and small-cap stocks. Logarithm of market capitalization	Captures the return differences between large-cap stocks and small-cap stocks Log of Market Capitalization

CNE5 Model Industry Factors	China A Long-Term Equity Model Industry Factors
1. Aerospace and Defense	1. Aerospace and Defense
2. Airlines	2. Airlines
3. Automobiles and Components	3. Apparel
4. Banks	4. Automobiles and Components
5. Beverages	5. Banks
6. Building Products	6. Beverages and Tobacco
7. Chemicals	7. Building Products
8. Commercial and Professional Services	8. Commercial and Professional Services
9. Construction and Engineering	9. Commodity Chemicals
10. Construction Materials	10. Conglomerates and Distributors
11. Diversified Financial Services	11. Construction and Engineering
12. Diversified Metals	12. Construction Materials
13. Electrical Equipment	13. Consumer Durables
14. Energy	14. Consumer Services
15. Food Products	15. Diversified Financials and Insurance
16. Food Staples Retail Household Personal Prod	16. Electrical Equipment
17. Hardware and Semiconductors	17. Energy
18. Health	18. Food Products
19. Hotels Restaurants and Leisure	19. Food Staples Retail and Household Personal Prod
20. Household Durables (non-Homebuilding)	20. Hardware and Semiconductors
21. Industrial Conglomerates	21. Health Care
22. Industrial Machinery	22. Machinery
23. Leisure Products Textiles Apparel and Luxury	23. Marine
24. Marine	24. Media
25. Materials	25. Metals and Mining

CNE5 Model Industry Factors	China A Long-Term Equity Model Industry Factors
26. Media	26. Non-Commodity Chemicals
27. Real Estate	27. Packaging and Paper and Forest Products
28. Retail	28. Real Estate
29. Road Rail and Transportation Infrastructure	29. Retailing
30. Software	30. Road and Rail Transportation Infrastructure
31. Trading Companies and Distributors	31. Software and Services
32. Utilities	32. Utilities

CNE5 Model Market Factor	China A Long-Term Equity Model Market Factor
China Country	China Country

CONTACT US

AMERICAS

Americas	1 888 588 4567 *
Atlanta	+ 1 404 551 3212
Boston	+ 1 617 532 0920
Chicago	+ 1 312 675 0545
Monterrey	+ 52 81 1253 4020
New York	+ 1 212 804 3901
San Francisco	+ 1 415 836 8800
São Paulo	+ 55 11 3706 1360
Toronto	+ 1 416 628 1007

EUROPE, MIDDLE EAST & AFRICA

Cape Town	+ 27 21 673 0100
Frankfurt	+ 49 69 133 859 00
Geneva	+ 41 22 817 9777
London	+ 44 20 7618 2222
Milan	+ 39 02 5849 0415
Paris	0800 91 59 17 *

ASIA PACIFIC

China North	10800 852 1032 *
China South	10800 152 1032 *
Hong Kong	+ 852 2844 9333
Mumbai	+ 91 22 6784 9160
Seoul	00798 8521 3392 *
Singapore	800 852 3749 *
Sydney	+ 61 2 9033 9333
Taipei	008 0112 7513 *
Thailand	0018 0015 6207 7181 *
Tokyo	+ 81 3 5290 1555

* = toll free

ABOUT MSCI

For more than 40 years, MSCI's research-based indexes and analytics have helped the world's leading investors build and manage better portfolios. Clients rely on our offerings for deeper insights into the drivers of performance and risk in their portfolios, broad asset class coverage, and innovative research.

Our line of products and services includes indexes, analytical models, data, real estate benchmarks, and ESG research.

MSCI serves 99 of the top 100 largest money managers, according to the most recent P&I ranking.

For more information, visit us at www.msci.com.

NOTICE AND DISCLAIMER

This document and all of the information contained in it, including without limitation all text, data, graphs, charts (collectively, the "Information") is the property of MSCI Inc. or its subsidiaries (collectively, "MSCI"), or MSCI's licensors, direct or indirect suppliers or any third party involved in making or compiling any Information (collectively, with MSCI, the "Information Providers") and is provided for informational purposes only. The Information may not be modified, reverse-engineered, reproduced or disseminated in whole or in part without prior written permission from MSCI.

The Information may not be used to create derivative works or to verify or correct other data or information. For example (but without limitation), the Information may not be used to create indexes, databases, risk models, analytics, software, or in connection with the issuing, offering, sponsoring, managing or marketing of any securities, portfolios, financial products or other investment vehicles utilizing or based on, linked to, tracking or otherwise derived from the Information or any other MSCI data, information, products or services.

The user of the Information assumes the entire risk of any use it may make or permit to be made of the Information. NONE OF THE INFORMATION PROVIDERS MAKES ANY EXPRESS OR IMPLIED WARRANTIES OR REPRESENTATIONS WITH RESPECT TO THE INFORMATION (OR THE RESULTS TO BE OBTAINED BY THE USE THEREOF), AND TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, EACH INFORMATION PROVIDER EXPRESSLY DISCLAIMS ALL IMPLIED WARRANTIES (INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTIES OF ORIGINALITY, ACCURACY, TIMELINESS, NON-INFRINGEMENT, COMPLETENESS, MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE) WITH RESPECT TO ANY OF THE INFORMATION.

Without limiting any of the foregoing and to the maximum extent permitted by applicable law, in no event shall any Information Provider have any liability regarding any of the Information for any direct, indirect, special, punitive, consequential (including lost profits) or any other damages even if notified of the possibility of such damages. The foregoing shall not exclude or limit any liability that may not by applicable law be excluded or limited, including without limitation (as applicable), any liability for death or personal injury to the extent that such injury results from the negligence or willful default of itself, its servants, agents or sub-contractors.

Information containing any historical information, data or analysis should not be taken as an indication or guarantee of any future performance, analysis, forecast or prediction. Past performance does not guarantee future results.

The Information should not be relied on and is not a substitute for the skill, judgment and experience of the user, its management, employees, advisors and/or clients when making investment and other business decisions. All Information is impersonal and not tailored to the needs of any person, entity or group of persons.

None of the Information constitutes an offer to sell (or a solicitation of an offer to buy), any security, financial product or other investment vehicle or any trading strategy.

It is not possible to invest directly in an index. Exposure to an asset class or trading strategy or other category represented by an index is only available through third party investable instruments (if any) based on that index. MSCI does not issue, sponsor, endorse, market, offer, review or otherwise express any opinion regarding any fund, ETF, derivative or other security, investment, financial product or trading strategy that is based on, linked to or seeks to provide an investment return related to the performance of any MSCI index (collectively, "Index Linked Investments"). MSCI makes no assurance that any Index Linked Investments will accurately track index performance or provide positive investment returns. MSCI Inc. is not an investment adviser or fiduciary and MSCI makes no representation regarding the advisability of investing in any Index Linked Investments.

Index returns do not represent the results of actual trading of investable assets/securities. MSCI maintains and calculates indexes, but does not manage actual assets. Index returns do not reflect payment of any sales charges or fees an investor may pay to purchase the securities underlying the index or Index Linked Investments. The imposition of these fees and charges would cause the performance of an Index Linked Investment to be different than the MSCI index performance.

The Information may contain back tested data. Back-tested performance is not actual performance, but is hypothetical. There are frequently material differences between back tested performance results and actual results subsequently achieved by any investment strategy.

Constituents of MSCI equity indexes are listed companies, which are included in or excluded from the indexes according to the application of the relevant index methodologies. Accordingly, constituents in MSCI equity indexes may include MSCI Inc., clients of MSCI or suppliers to MSCI. Inclusion of a security within an MSCI index is not a recommendation by MSCI to buy, sell, or hold such security, nor is it considered to be investment advice.

Data and information produced by various affiliates of MSCI Inc., including MSCI ESG Research LLC and Barra LLC, may be used in calculating certain MSCI indexes. More information can be found in the relevant index methodologies on www.msci.com.

MSCI receives compensation in connection with licensing its indexes to third parties. MSCI Inc.'s revenue includes fees based on assets in Index Linked Investments. Information can be found in MSCI Inc.'s company filings on the Investor Relations section of www.msci.com.

MSCI ESG Research LLC is a Registered Investment Adviser under the Investment Advisers Act of 1940 and a subsidiary of MSCI Inc. Except with respect to any applicable products or services from MSCI ESG Research, neither MSCI nor any of its products or services recommends, endorses, approves or otherwise expresses any opinion regarding any issuer, securities, financial products or instruments or trading strategies and MSCI's products or services are not intended to constitute investment advice or a recommendation to make (or refrain from making) any kind of investment decision and may not be relied on as such. Issuers mentioned or included in any MSCI ESG Research materials may include MSCI Inc., clients of MSCI or suppliers to MSCI, and may also purchase research or other products or services from MSCI ESG Research. MSCI ESG Research materials, including materials utilized in any MSCI ESG Indexes or other products, have not been submitted to, nor received approval from, the United States Securities and Exchange Commission or any other regulatory body.

Any use of or access to products, services or information of MSCI requires a license from MSCI. MSCI, Barra, RiskMetrics, IPD, and other MSCI brands and product names are the trademarks, service marks, or registered trademarks of MSCI or its subsidiaries in the United States and other jurisdictions. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and Standard & Poor's. "Global Industry Classification Standard (GICS)" is a service mark of MSCI and Standard & Poor's.